

Day 5 Deliverable — Pipeline & Experiment Planning

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Dataset: LG HG2 18650 Li-ion Battery Dataset (McMaster University) | Organization: TIH IIT Guwahati | AI/ML Internship 2026

Part 1: Reproducible Pipeline Flowchart

The following pipeline describes every step from raw CSV files to final evaluated models. Each step is documented with inputs, outputs, and key decisions to ensure full reproducibility.

S1	Raw Data Ingestion Load all 208 CSV files across 6 temperature folders. Skip 28 metadata header rows. Parse Voltage, Current, Temperature, Capacity, WhAccu, Time Stamp, Status, and Step columns. Store parsed DataFrames in a dictionary keyed by temperature and filename.
S2	Data Cleaning Filter rows where Status = PAU (tester paused). Force numeric dtypes on all sensor columns. Drop rows with NaN in Voltage, Current, or Temperature. Check timestamp monotonicity per file. Flag and remove sensor spike outliers (Voltage outside 2.5–4.3V, Current outside –20A to +10A).
S3	SoC Label Derivation Derive SoC from Capacity column per continuous segment using Coulomb counting: $SoC(t) = SoC_initial + (Capacity_t / Q_nominal) \times 100$. Use temperature-specific $Q_nominal$ from Cap_1C files. Assume SoC = 100% immediately after a completed charge cycle. Assume SoC = 0% at end of a full discharge. Stitch segments where start SoC is known.
S4	Remaining Charging Time Label Derivation For Charge files only: compute Remaining Charging Time at each row as $timestamp_end_of_charge - timestamp_current$. Define end of charge as the row where Current first drops below 150mA (C/20 threshold) in the CV phase. Exclude Charge files with fewer than 500 rows.
S5	Exploratory Data Analysis (EDA) Plot voltage, current, and temperature vs time for 3 representative files per temperature. Plot SoC histogram across all training files. Plot correlation heatmap of all features vs SoC. Analyze CC vs CV phase distribution. Identify files with anomalous patterns. Document all findings.
S6	Feature Engineering Create derived features per file (no cross-file rolling): rolling_mean_voltage (window 10), rolling_mean_current (window 10), dV_dt (rate of change of voltage per timestep), power (Voltage $\times$ Current), time_elapsed (seconds from file start), cc_cv_flag (1 if Current < 0.5A during charging = CV phase, else 0). Add temperature_folder as a category feature.
S7	Train / Validation / Test Split Split by file — never by row. Training: Mixed1–6 from all 6 temperature folders + Charge1–12. Validation: Mixed7 from all temperatures + Charge13–14. Test: Mixed8, UDDS, HWFET, LA92, US06 from all temperatures + Charge15–16. No row shuffling. Each file stays entirely in one split.
S8	Feature Scaling Fit StandardScaler on training data only. Apply the same fitted scaler to validation and test sets. Save the fitted scaler object for reproducibility. Do NOT fit scaler on validation or test data.
S9	Model Training Train 4 models in sequence: (1) Linear Regression — no hyperparameters, (2) Random Forest — n_estimators=100, max_depth=None, random_state=42, (3) XGBoost — n_estimators=500, learning_rate=0.05, max_depth=6, subsample=0.8, early_stopping_rounds=20, (4) ANN — Dense(64)→Dense(32)→Dense(1), Adam, MSE loss, early stopping patience=10. Train separately for SoC target and Remaining Charging Time target.
S10	Evaluation Evaluate each model on validation set first. Report MAE, RMSE, and R-squared separately for SoC and Remaining Charging Time. Evaluate final best model on test set. Report per-temperature performance. Report per-SoC-range performance (0–20%, 20–80%, 80–100%).
S11	Error and Feature Importance Analysis Plot residuals vs SoC range. Plot residuals for CC phase vs CV phase separately. Extract feature importance from RF and XGBoost. Plot top 10 features. Identify which temperature condition gives worst performance.
S12	Final Report and Submission Compile all results into comparison table. Write final report. Save trained model weights (joblib for RF/XGBoost, .h5 for ANN). Push all notebooks, results, and model files to GitHub.

Part 2: Experiment Plan

2.1 Baseline Experiment

The baseline establishes the minimum acceptable performance and quantifies how much complex models add over a simple approach.

Parameter	Details
Experiment ID	EXP-001
Model	Linear Regression (sklearn.linear_model.LinearRegression)
Target 1	SoC (%) — Coulomb counting derived label
Target 2	Remaining Charging Time (minutes) — from Charge files only
Input Features	Voltage, Current, Temperature, Capacity, time_elapsed, power
Preprocessing	StandardScaler on training data. No rolling features for baseline.
Training files	Mixed1–6 all temperatures (SoC); Charge1–12 all temperatures (RCT)
Validation files	Mixed7 all temperatures (SoC); Charge13–14 (RCT)
Expected SoC result	$R^2 \sim 0.70\text{--}0.80$, MAE $\sim 5\text{--}10\%$ SoC based on literature
Expected RCT result	$R^2 \sim 0.60\text{--}0.75$, MAE $\sim 8\text{--}15$ minutes
Purpose	Set performance floor. Quantify the improvement from non-linear models.

2.2 Advanced Experiment 1 — Random Forest

Parameter	Details
Experiment ID	EXP-002
Model	Random Forest Regressor (sklearn.ensemble.RandomForestRegressor)
Hyperparameters	n_estimators=100, max_depth=None, min_samples_leaf=2, random_state=42, n_jobs=-1
Input Features	Voltage, Current, Temperature, Capacity, power, time_elapsed, rolling_mean_voltage, rolling_mean_current, dV_dt, cc_cv_flag
Additional steps	Extract feature importances. Retrain with top 7 features only and compare.
Expected SoC result	$R^2 > 0.92$, MAE $< 3\%$ SoC
Expected RCT result	$R^2 > 0.88$, MAE < 5 minutes
Key advantage	Feature importance scores directly interpretable. Robust to outliers and temperature variation.

2.3 Advanced Experiment 2 — XGBoost

Parameter	Details
Experiment ID	EXP-003
Model	XGBoost Regressor (xgboost.XGBRegressor)
Hyperparameters	n_estimators=500, learning_rate=0.05, max_depth=6, subsample=0.8, colsample_bytree=0.8, reg_alpha=0.1, reg_lambda=1.0, early_stopping_rounds=20, random_state=42
Input Features	Same as EXP-002 plus WhAccu as additional energy feature
Early stopping	Monitor validation RMSE. Stop if no improvement for 20 rounds.
Expected SoC result	$R^2 > 0.96$, MAE $< 2\%$ SoC
Expected RCT result	$R^2 > 0.92$, MAE < 3.5 minutes
Key advantage	Built-in regularization. Fastest inference. Best non-deep-learning performance per literature.

2.4 Project-Specific Experiment — Temperature Generalization Study

This experiment is specific to our LG HG2 dataset and goes beyond what a standard ML tutorial would include. It tests whether models trained on one temperature can generalize to other temperatures.

Parameter	Details
Experiment ID	EXP-004
Question	Does including all 6 temperatures in training significantly outperform training only at 25°C?
Setup A	Train XGBoost on 25°C Mixed files only. Test on all 6 temperatures separately.
Setup B	Train XGBoost on Mixed files from all 6 temperatures. Test on all 6 temperatures separately.

Parameter	Details
Metric	Per-temperature MAE and RMSE for SoC. Compare Setup A vs Setup B for each temperature.
Expected finding	Setup B (multi-temperature training) should significantly outperform Setup A at extreme temperatures (−20°C, −10°C, 40°C).
Why it matters	Real EVs operate across a wide temperature range. A model trained only at room temperature would fail in winter or hot climates.

2.5 Evaluation Criteria

All experiments are evaluated using the following criteria, reported separately for SoC prediction and Remaining Charging Time prediction:

Metric	Formula	Target (SoC)	Target (RCT)
MAE	$(1/n) \times \sum y - \hat{y} $	< 3% SoC (good), < 1.5% (excellent)	< 5 min (good), < 2.5 min (excellent)
RMSE	$\sqrt{(1/n) \times \sum (y - \hat{y})^2}$	< 4% SoC (good), < 2% (excellent)	< 7 min (good), < 3.5 min (excellent)
R-squared	$1 - (SS_{res} / SS_{tot})$	> 0.90 (good), > 0.96 (excellent)	> 0.85 (good), > 0.92 (excellent)

In addition, per-temperature breakdown and per-SoC-range breakdown (0–20%, 20–80%, 80–100%) will be reported for the best model to identify where predictions are weakest.

Part 3: Experiment Log Template

Every experiment run must be logged using the following template. One entry per experiment per run. Logs are stored in results/experiment_log.csv in the repository.

3.1 Log Fields

Field	Type	Description
exp_id	String	Unique experiment ID — format: EXP-XXX (e.g., EXP-001)
date	Date	Date the experiment was run — format: YYYY-MM-DD
model	String	Model name: LinearRegression / RandomForest / XGBoost / ANN
target	String	SoC / RemainingChargingTime
features	String	Comma-separated list of input features used
train_files	String	Description of training files used (e.g., Mixed1-6 all temps)
val_files	String	Validation files used
hyperparameters	JSON String	Key hyperparameters as a JSON string (e.g., {"n_estimators": 100})
val_mae	Float	Mean Absolute Error on validation set
val_rmse	Float	Root Mean Squared Error on validation set
val_r2	Float	R-squared on validation set
test_mae	Float	MAE on test set — only filled for the final selected model
test_rmse	Float	RMSE on test set — only filled for the final selected model
test_r2	Float	R-squared on test set — only filled for the final selected model
notes	String	Any observations, anomalies, or changes made during this run
notebook_ref	String	Filename of the notebook that produced this result (e.g., Day6_Baseline.ipynb)

3.2 Sample Log Entry (filled)

Field	Value
exp_id	EXP-001
date	2026-08-25
model	LinearRegression
target	SoC
features	Voltage, Current, Temperature, Capacity, power, time_elapsed

Field	Value
train_files	Mixed1-6 all 6 temperature folders
val_files	Mixed7 all 6 temperature folders
hyperparameters	{ } (no hyperparameters)
val_mae	6.84
val_rmse	9.21
val_r2	0.743
test_mae	— (not yet evaluated)
test_rmse	— (not yet evaluated)
test_r2	— (not yet evaluated)
notes	Baseline run. Model underfits near 0% and 100% SoC as expected. Rolling features not included yet.
notebook_ref	WEEK_1_Day_6_SreetamaSantra.ipynb

3.3 CSV Header (for experiment_log.csv)

The following header line should be the first row of results/experiment_log.csv:

exp_id,date,model,target,features,train_files,val_files,hyperparameters,val_mae,val_rmse,val_r2,test_mae,test_rmse,test_r2,notes,notebook_ref

Part 4: Repository and Folder Structure

The following structure is defined for the GitHub repository TIH-SreetamaSantra. All files must follow the naming convention WEEK_x_Day_y_SreetamaSantra.

Path	Type	Contents
TIH-SreetamaSantra/	Root	GitHub repository root
■■■■ README.md	File	Project overview, dataset description, file naming guide
■■■■ Week1/	Folder	All Week 1 daily deliverables
■ ■■■■ WEEK_1_Day_1_SreetamaSantra.pdf	PDF	Problem Understanding Note, Pipeline Diagram, Questions List
■ ■■■■ WEEK_1_Day_2_SreetamaSantra.pdf	PDF	Literature Review, Comparison Table, Candidate Method List
■ ■■■■ WEEK_1_Day_3_SreetamaSantra.pptx	PPT	Mentor Review 1 — Progress, Findings, Blockers, Next Steps
■ ■■■■ WEEK_1_Day_4_SreetamaSantra.pdf	PDF	Dataset Schema Note and Risk Checklist
■ ■■■■ WEEK_1_Day_5_SreetamaSantra.pdf	PDF	Pipeline Flowchart, Experiment Plan, Log Template, Folder Structure
■ ■■■■ WEEK_1_Day_6_SreetamaSantra.ipynb	Notebook	Baseline Linear Regression model — SoC and RCT
■ ■■■■ WEEK_1_Day_7_SreetamaSantra.pdf	PDF	Intern Discussion notes and EDA comparison summary
■■■■ Week2/	Folder	Week 2 deliverables — EDA, Feature Engineering, RF model
■ ■■■■ WEEK_2_Day_8_SreetamaSantra.ipynb	Notebook	Full EDA — distributions, correlations, CC/CV phase analysis
■ ■■■■ WEEK_2_Day_9_SreetamaSantra.ipynb	Notebook	Feature engineering pipeline
■ ■■■■ WEEK_2_Day_10_SreetamaSantra.pptx	PPT	Mentor Review 2
■ ■■■■ ...		Remaining Week 2 files follow same pattern
■■■■ Week3/	Folder	XGBoost, ANN experiments and tuning
■■■■ Week4/	Folder	Final model, error analysis, report and presentation
■■■■ data/	Folder	NOT uploaded to GitHub (dataset is too large). Referenced via Drive link only.
■■■■ models/	Folder	Saved model files — .joblib for RF/XGBoost, .h5 for ANN
■ ■■■■ rf_soc_model.joblib	Model	Trained Random Forest for SoC
■ ■■■■ xgb_soc_model.joblib	Model	Trained XGBoost for SoC
■ ■■■■ rf_rct_model.joblib	Model	Trained Random Forest for Remaining Charging Time
■ ■■■■ scaler.joblib	Model	Fitted StandardScaler — must match training data

Path	Type	Contents
■■■■ results/	Folder	All evaluation outputs
■■■■ experiment_log.csv	CSV	One row per experiment run — see log template in Part 3
■■■■ figures/	Folder	All plots: EDA figures, training curves, residual plots, feature importance charts
■■■■ metrics_summary.csv	CSV	Final comparison table of all models across all metrics

4.1 Notebook Structure Convention

Every notebook must follow this internal structure to ensure reproducibility:

Section 0 — Imports and configuration (random seeds, file paths, constants)

Section 1 — Data loading and cleaning

Section 2 — Feature engineering (if applicable)

Section 3 — Model training

Section 4 — Evaluation and metrics

Section 5 — Plots and visualizations

Section 6 — Log entry — print the experiment_log row at the end of every notebook

The first cell of every notebook must set the random seed:

```
import numpy as np, random
```

```
np.random.seed(42)
```

```
random.seed(42)
```

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